

Machine Learning for Business

GTSC2143

Lecture 6 Classification I

Instructor: Dr. Lily Bei Luo

AY2025-2026 Semester 1

Case: Predicting Sentiment by Topic –
An intelligent restaurant review system

Predicting sentiment by topic

It's a big day & I want to book a table at a nice Japanese restaurant



Predicting sentiment by topic



Sample review:

Watching the chefs create incredible edible art made the experience very unique.

My wife tried their ramen and it was pretty forgettable.

All the sushi was delicious!
Easily best sushi in Seattle.

Positive reviews not positive about everything

Experience



Predicting sentiment by topic

From reviews to topic sentiments

All reviews for restaurant

★★★★☆ 7/21/2015

This is probably my favorite place to eat Japanese in Seattle. My boyfriend and I ordered nigiri of scallop, Japanese snapper (seasonal), and the agedashi tofu and 2 special rolls. I would skip the special rolls, because the nigiri and sashimi cuts is where this place excels. The tofu, as recommended by other Yelpers was amazing. It's more chewy and the sauce/gravy is the perfect amount of flavor for the delicate tofu.

★★★★★ 6/11/2015

Dining here at the sushi bar made me feel like sitting front row to an amazing performance. We didn't have resos, banged down to the ID after work, got here breathlessly at 5:10pm, and got the last two seats in the place.

★★★★☆ 6/9/2015

I came here having high expectations due to the reviews of this place, but I was bit disappointed. The restaurant is small so do make reservations when you come here. Dishes cost from \$4-26 each and dishes are small.

Novel intelligent
restaurant review app

Experience



Ramen



Sushi



Easily best sushi
in Seattle.

Predicting sentiment by topic

Intelligent restaurant review system

All reviews for restaurant

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Break all reviews into sentences

The seaweed salad was just OK, vegetable
salad was just ordinary.

I like the interior decoration and the
blackboard menu on the wall.

All the sushi was delicious.

My wife tried their ramen and it was pretty
forgettable.

The sushi was amazing, and the rice is just
outstanding.

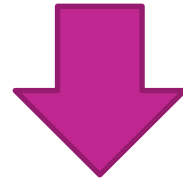
The service is somewhat hectic.

Easily best sushi in Seattle.

Predicting sentiment by topic

Core building block

Easily best sushi in Seattle.



Sentence Sentiment
Classifier



Easily best sushi in Seattle.



Predicting sentiment by topic

Intelligent restaurant review system

All reviews for restaurant

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Easily best sushi in Seattle.

Sentence
Sentiment
Classifier

Average
predictions

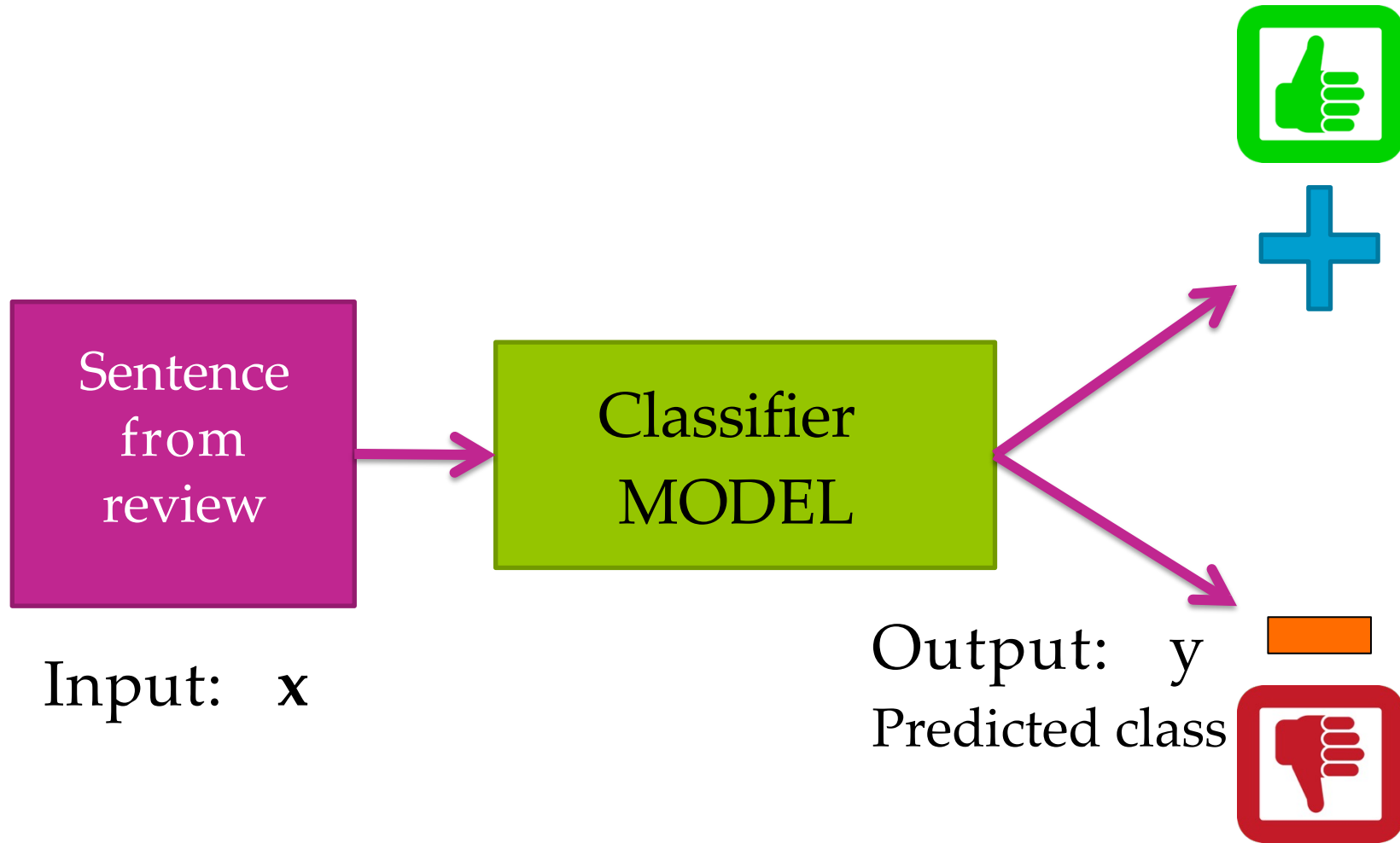
Sushi
★★★★★

Most
👍 & 👎

Easily best
sushi
in Seattle.

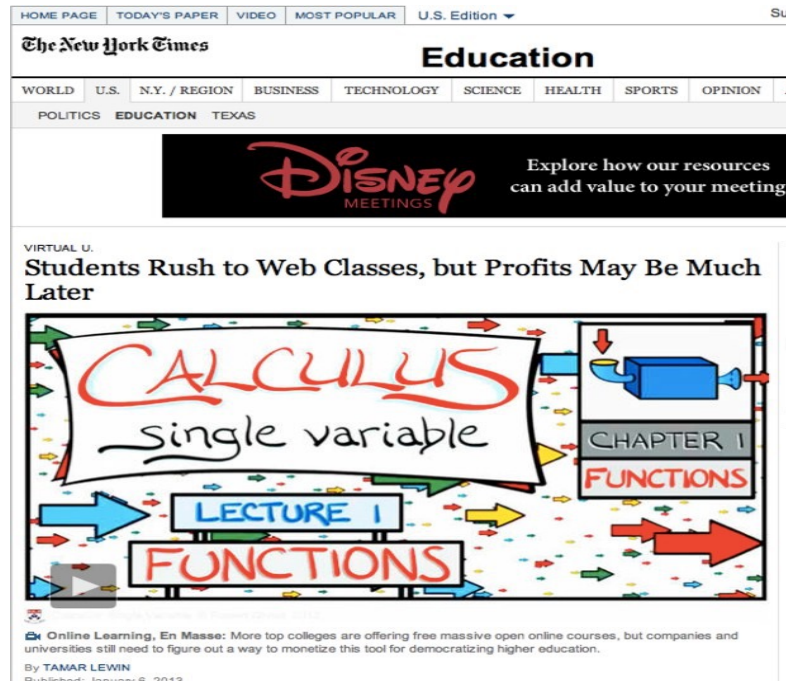
Classifier Applications

Classifier

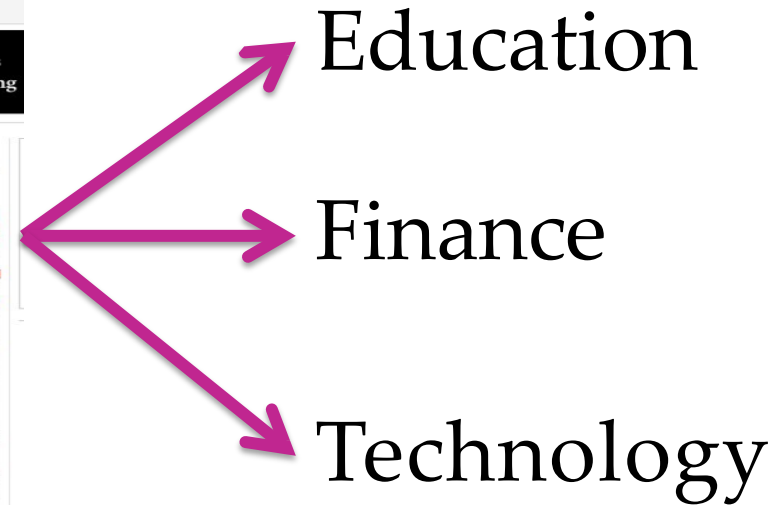


Example multiclass classifier

What's type of this article?



Input: x
Webpage



Output: y

Spam filtering

Is this email spam?

★ **Osman Khan** to Carlos [show details](#) Jan 7 (6 days ago) [Reply](#) ▼

sounds good
+ok

Carlos Guestrin wrote:
Let's try to chat on Friday a little to coordinate and more on Sunday in person?

Carlos

Welcome to New Media Installation: Art that Learns

★ **Carlos Guestrin** to 10615-announce, Osman, Miche [show details](#) 3:15 PM (8 hours ago) [Reply](#) ▼

Hi everyone,

Welcome to New Media Installation: Art that Learns

The class will start tomorrow.
Make sure you attend the first class, even if you are on the Wait List.
The classes are held in Doherty Hall C316, and will be Tue, Thu 01:30-4:20 PM.

By now, you should be subscribed to our course mailing list: 10615-announce@cs.cmu.edu.
You can contact the instructors by emailing: 10615-instructors@cs.cmu.edu

Natural _LoseWeight SuperFood Endorsed by Oprah Winfrey, Free Trial 1 bottle, pay only \$5.95 for shipping mfw rik [Spam](#) | X

★ **Jaquelyn Halley** to nherlein, bcc: thehorney, bcc: anç [show details](#) 9:52 PM (1 hour ago) [Reply](#) ▼

=== Natural WeightLOSS Solution ===

Vital Acai is a natural WeightLOSS product that Enables people to lose wieght and cleansing their bodies faster than most other products on the market.

Here are some of the benefits of Vital Acai that You might not be aware of. These benefits have helped people who have been using Vital Acai daily to Achieve goals and reach new heights in there dieting that they never thought they could.

- * Rapid WeightLOSS
- * Increased metabolism - BurnFat & calories easily!
- * Better Mood and Attitude
- * More Self Confidence
- * Cleanse and Detoxify Your Body
- * Much More Energy

Not spam

Spam

Input: x

Output: y

Text of email,
sender, IP,...

Image classification



Input: x
Image pixels



Output: y
Predicted object

Personalized medical diagnosis

Input: x



Output: y

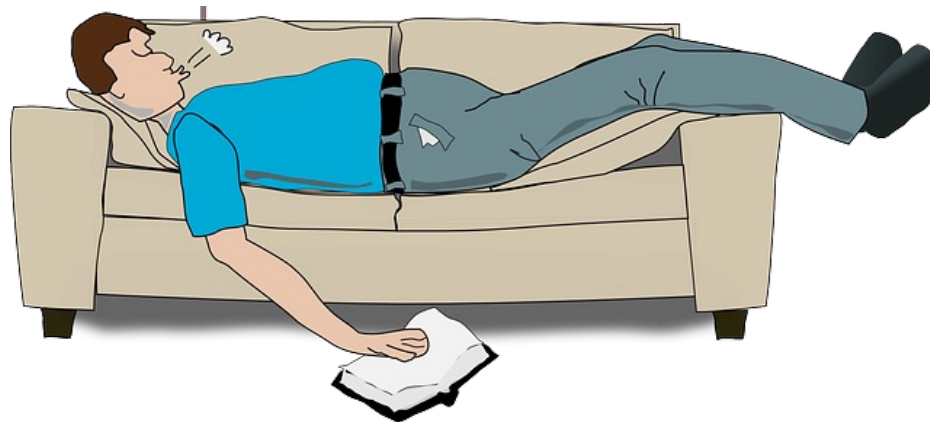
Disease
Classifier
MODEL

Healthy

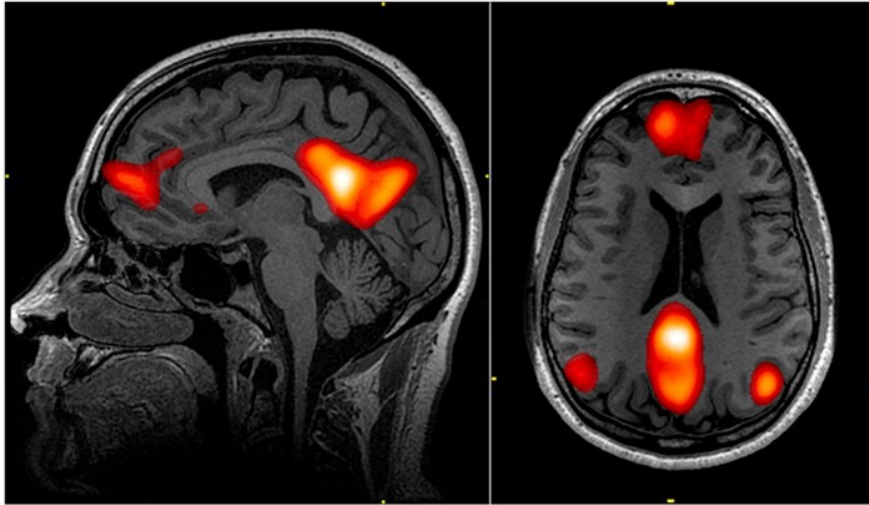
Cold Flu

Pneumonia

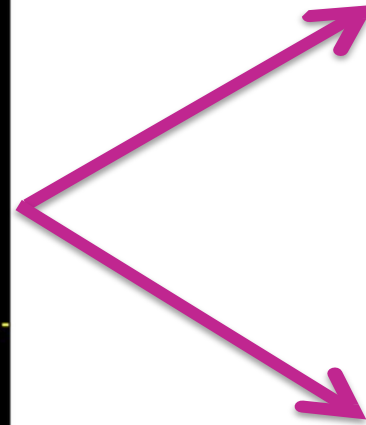
...



Reading your mind



“Hammer”

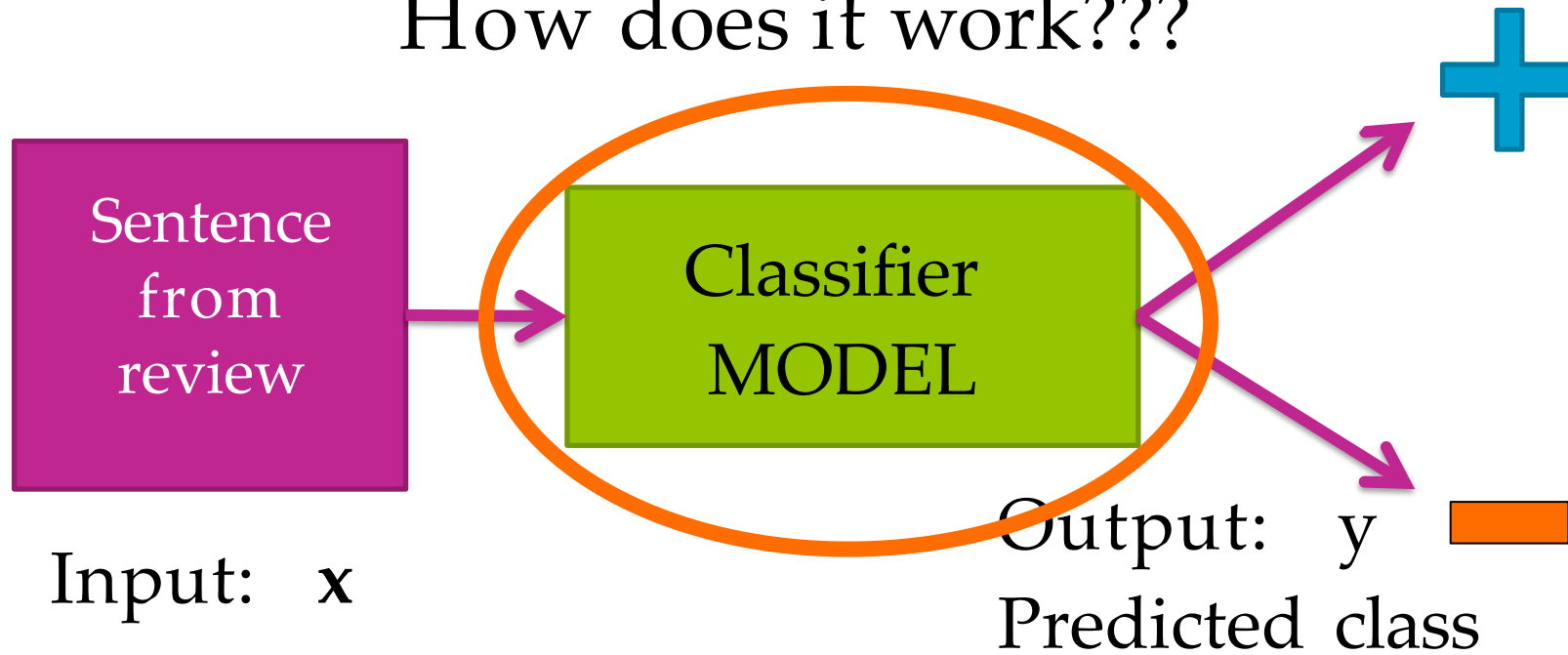


“House”

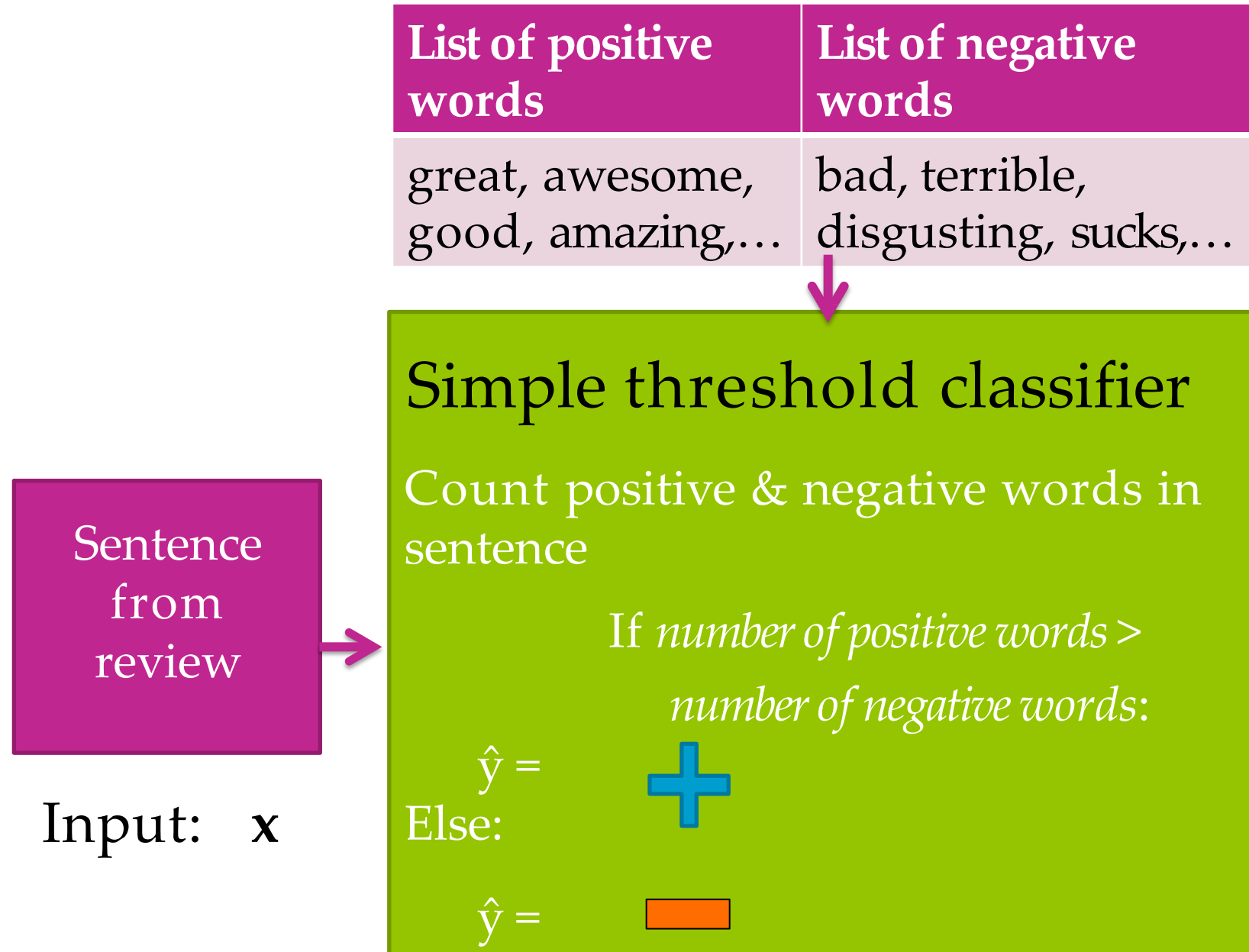
Linear Classifiers

Representing classifiers

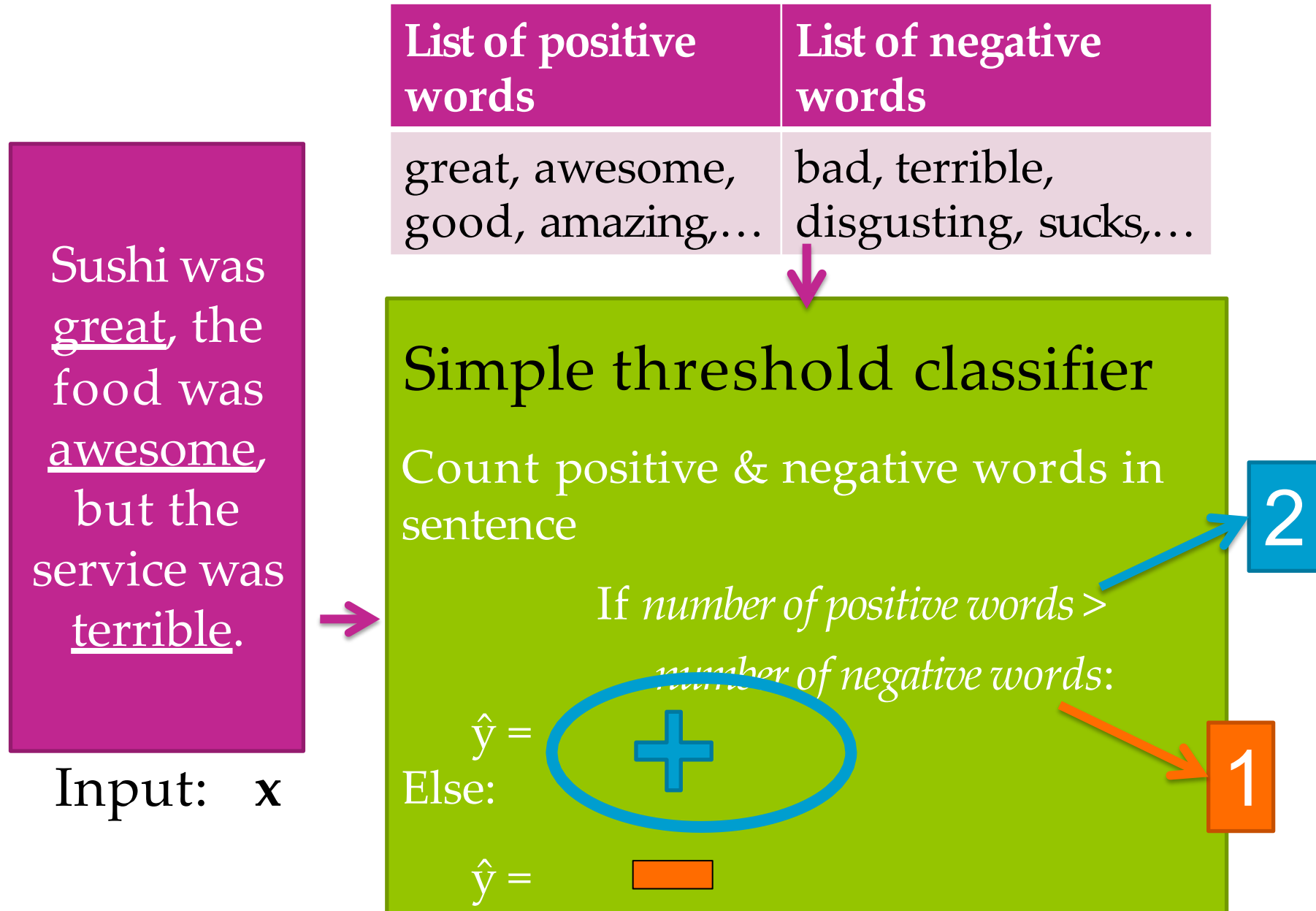
How does it work???



Representing classifiers



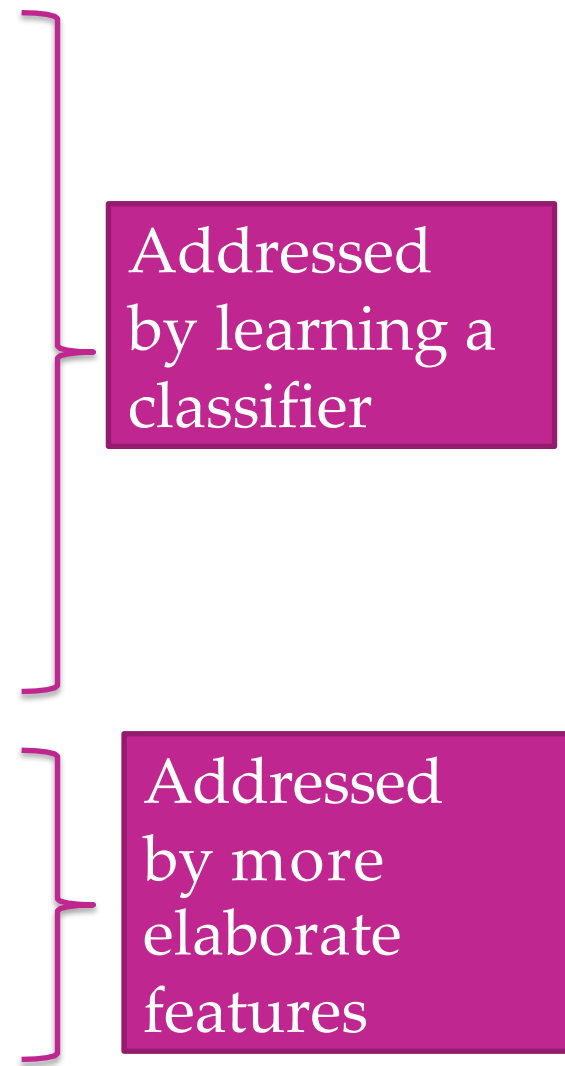
Representing classifiers



Representing classifiers

Problems with threshold classifier

- How do we get list of positive/negative words?
- Words have different degrees of sentiment:
 - Great > good
 - How do we weigh different words?
- Single words are not enough:
 - *Good* → Positive
 - *Not good* → Negative



Addressed
by learning a
classifier

Addressed
by more
elaborate
features

Representing classifiers

A (linear) classifier

- Will use training data to learn a weight for each word

Word	Weight
good	1.0
great	1.5
awesome	2.7
bad	-1.0
terrible	-2.1
awful	-3.3
restaurant, the, we, where, ...	0.0
...	...

Representing classifiers

Scoring a sentence

Word	Weight
good	1.0
great	1.2
awesome	1.7
bad	-1.0
terrible	-2.1
awful	-3.3
restaurant, the, we, where, ...	0.0
...	...

Input x:

Sushi was great,
the food was awesome,
but the service was terrible.

Called a linear classifier, because output is weighted sum of input.

Representing classifiers

Word	Weight
...	...



Simple linear classifier

$Score(x)$ = weighted count of words in sentence

Sentence
from
review



Input: x

If $Score(x) > 0$:

$\hat{y} =$ 

Else:

$\hat{y} =$ 

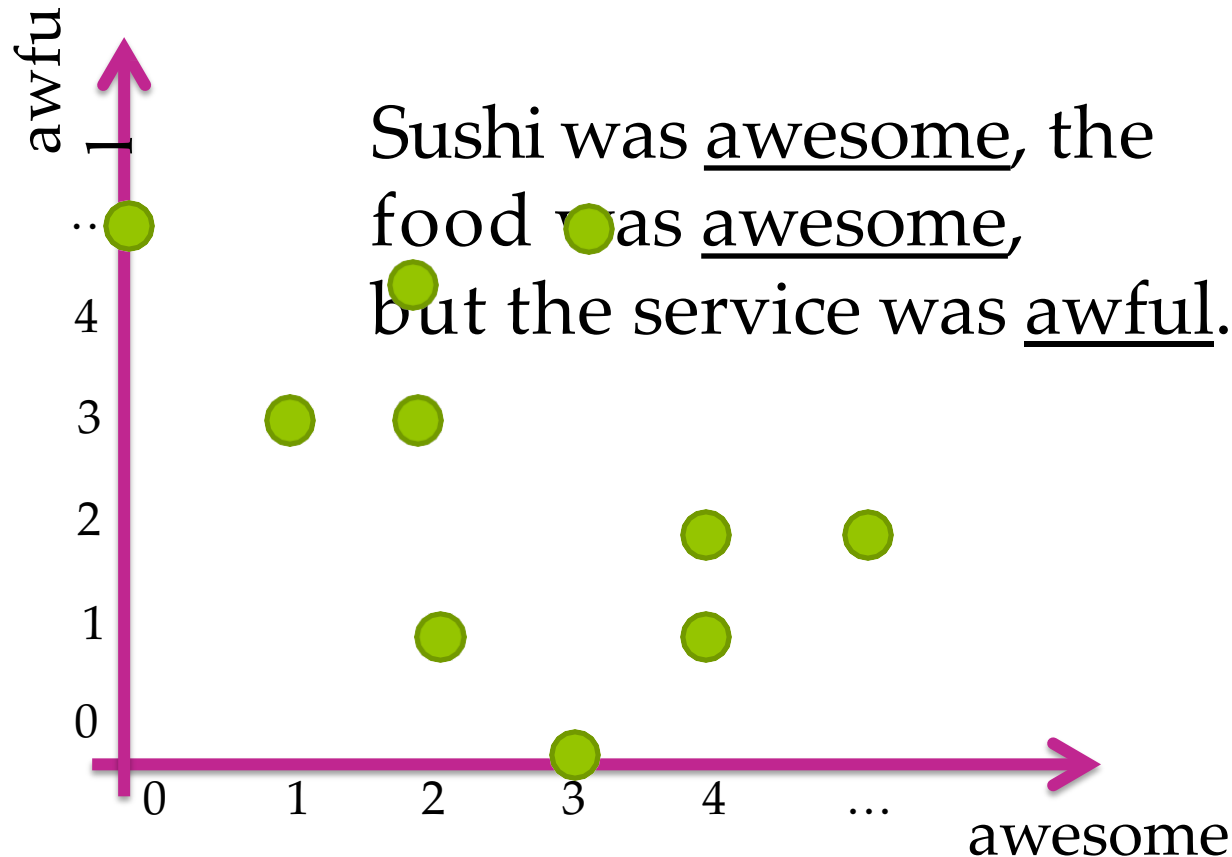
Decision Boundaries

Decision boundaries

Suppose only two words had non-zero weight

Word	Weight
awesome	1.0
awful	-1.5

➔ $\text{Score}(x) = 1.0 \# \text{awesome} - 1.5 \# \text{awful}$

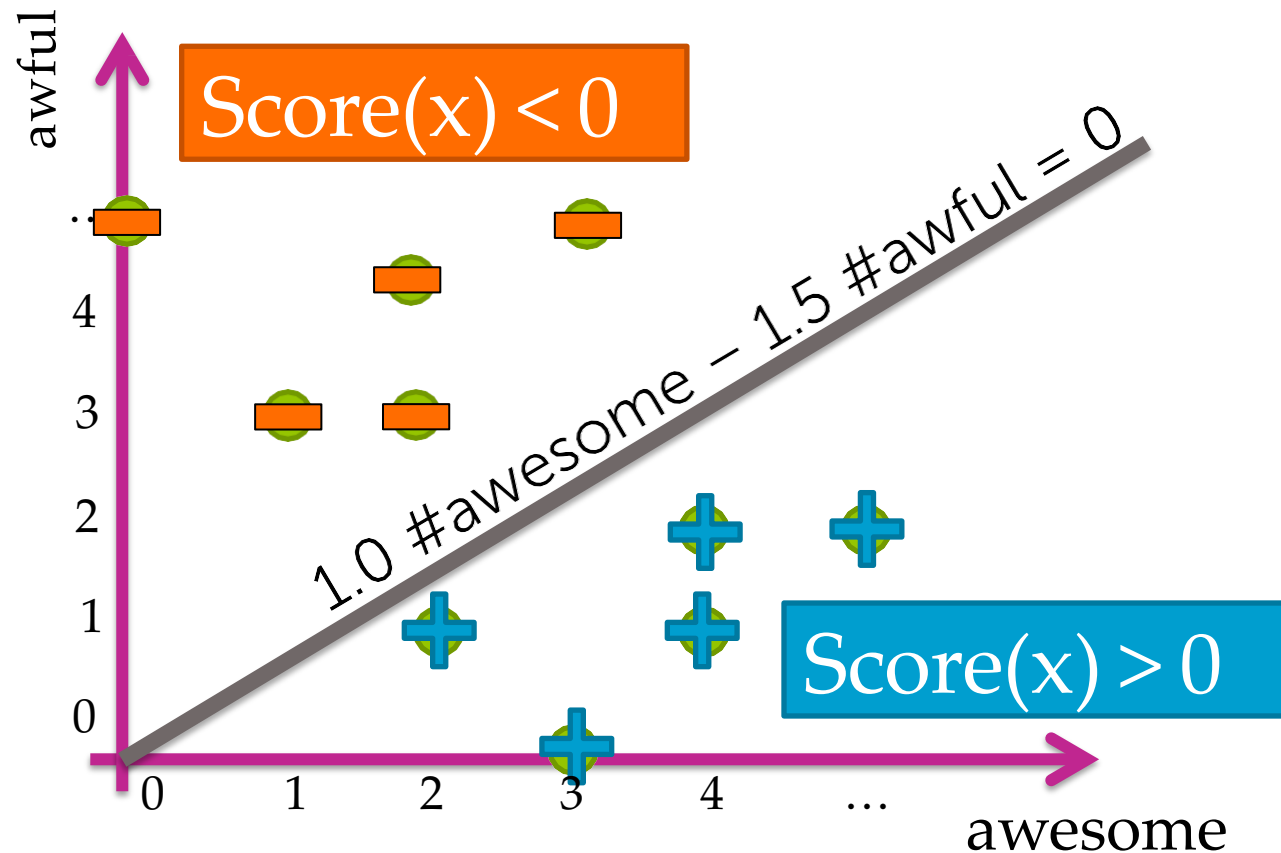


Decision boundaries

Suppose only two words had non-zero weight

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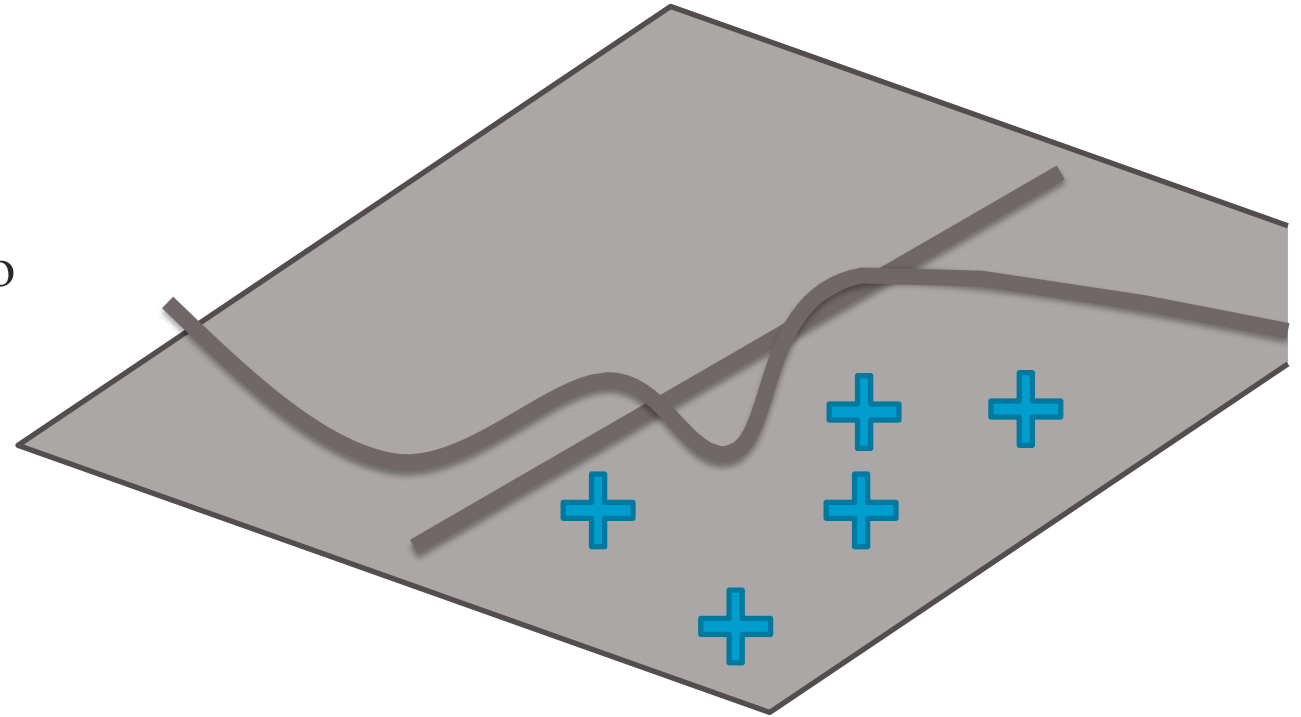
➔ $\text{Score}(x) = 1.0 \# \text{awesome} - 1.5 \# \text{awful}$



Decision boundaries

Decision boundary separates positive & negative predictions

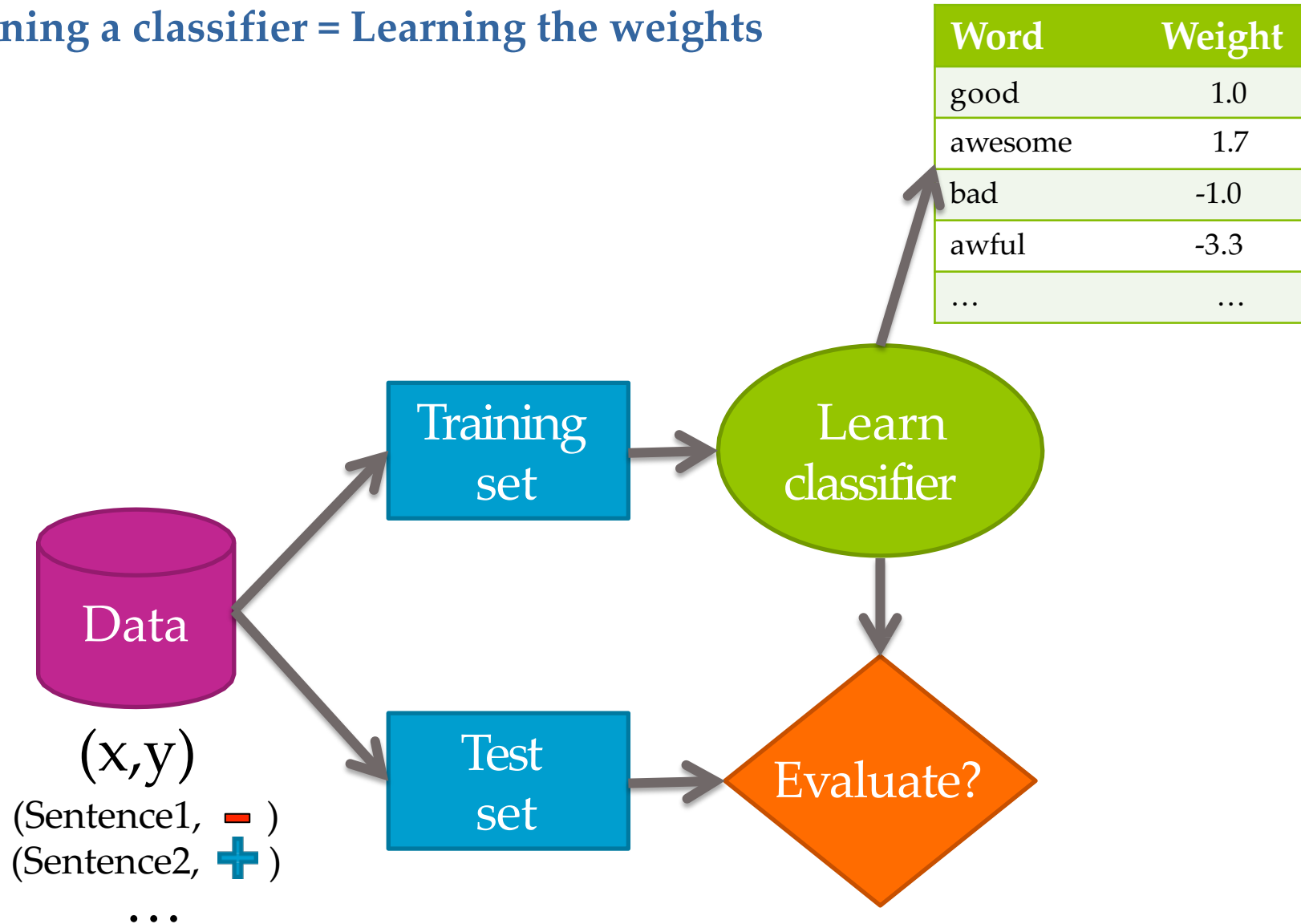
- For linear classifiers:
 - When 2 weights are non-zero
→ **line**
 - When 3 weights are non-zero
→ **plane**
 - When many weights are non-zero
→ **hyperplane**
- For more general classifiers
→ **more complicated shapes**



Training and Evaluating a Classifier

Training and Evaluating a Classifier

Training a classifier = Learning the weights



Training and Evaluating a Classifier

Classification Error & Accuracy

- Error measures fraction of mistakes

$$\text{error} = \frac{\text{\# of mistakes}}{\text{Total \# of sentences}}$$

- Best possible value is 0.0

- Often, measure **accuracy**

- Fraction of correct predictions

$$\text{accuracy} = \frac{\text{\# of correct}}{\text{Total \# of sentences}}$$

- Best possible value is 1.0

What's a good accuracy?

Training and Evaluating a Classifier

What if you ignore the sentence, and just guess?

- For binary classification:
 - Half the time, you'll get it right! (on average)
→ accuracy = 0.5
- For k classes, accuracy = $1/k$
 - 0.333 for 3 classes, 0.25 for 4 classes,...

At the very, very, very least,
you should healthily beat random... Otherwise, it's
(usually) pointless...

Training and Evaluating a Classifier

Is a classifier with 90% accuracy good? Depends...

2010 data shows:

"90% emails sent are spam!"

Predicting every email is spam
gets you 90% accuracy!!!

Majority class prediction

Amazing performance when
there is class imbalance

(but silly approach)

- One class is more common than others
- Beats random (if you know the majority class)

Training and Evaluating a Classifier ---

So, always be digging in and asking the hard questions about reported accuracies

- Is there class imbalance?
- How does it compare to a simple, baseline approach?
 - Random guessing
 - Majority class
 - ...
- Most importantly:
what accuracy does my application need?
 - What is good enough for my user's experience?
 - What is the impact of the mistakes we make?

Training and Evaluating a Classifier

Types of Mistakes

		Predicted label	
			
True label		True Positive	False Negative (FN)
		False Positive (FP)	True Negative

Training and Evaluating a Classifier

Cost of different types of mistakes can be different (& high) in some applications

	Spam filtering	Medical diagnosis
False negative	Annoying	Disease not treated
False positive	Email lost	Wasteful treatment

Training and Evaluating a Classifier

Confusion matrix - binary classification

			
True label		50	10
		5	35

Training and Evaluating a Classifier

Confusion matrix - multiclass classification

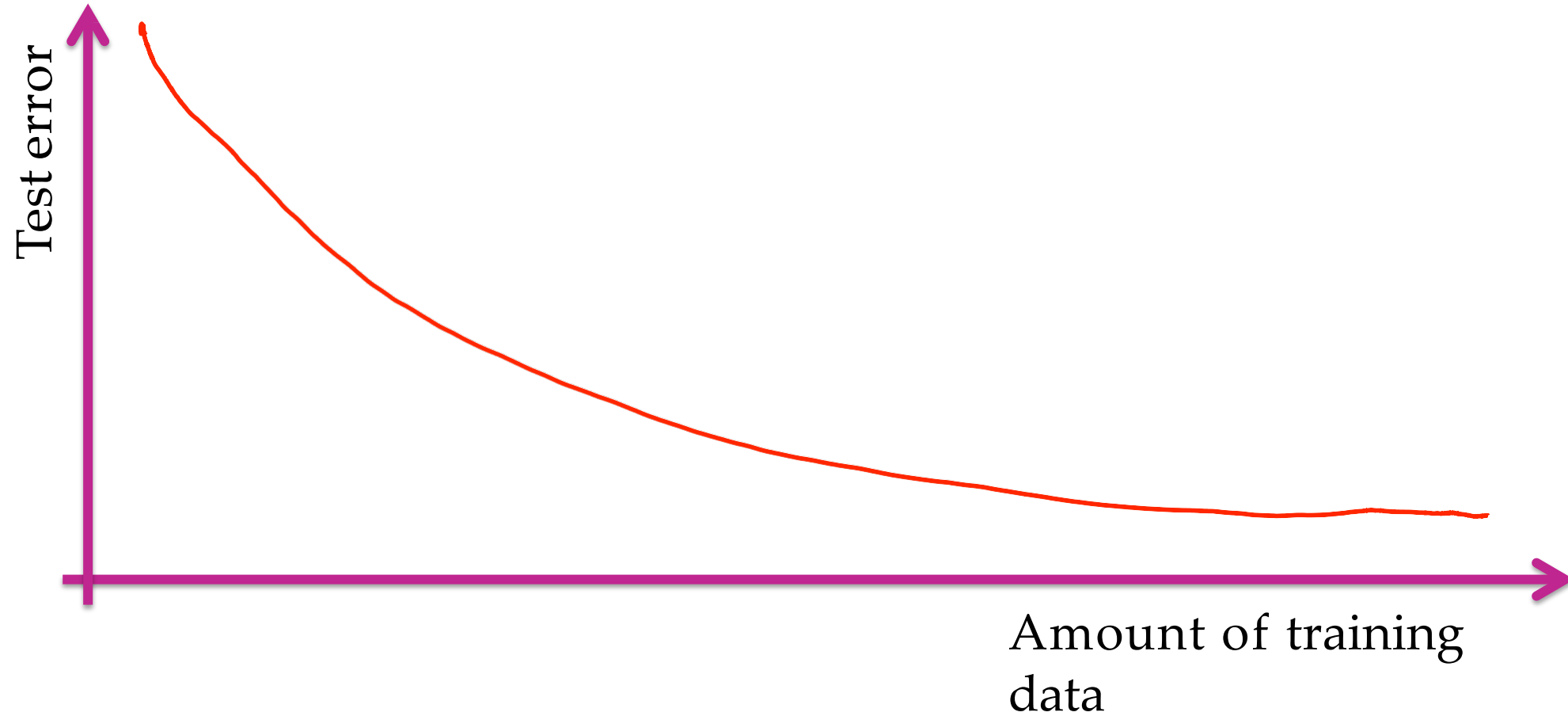
		Predicted label		
		Healthy	Cold	Flu
True label	Healthy 70	60	8	2
	Cold 20	4	12	4
	Flu 10	0	2	8

Learning curves: *How much data do I need?*

Learning curves

- The more the merrier
 - But data quality is most important factor
- Theoretical techniques sometimes can bound how much data is needed
 - Typically too loose for practical application
 - But provide guidance
- In practice:
 - More complex models require more data
 - Empirical analysis can provide guidance

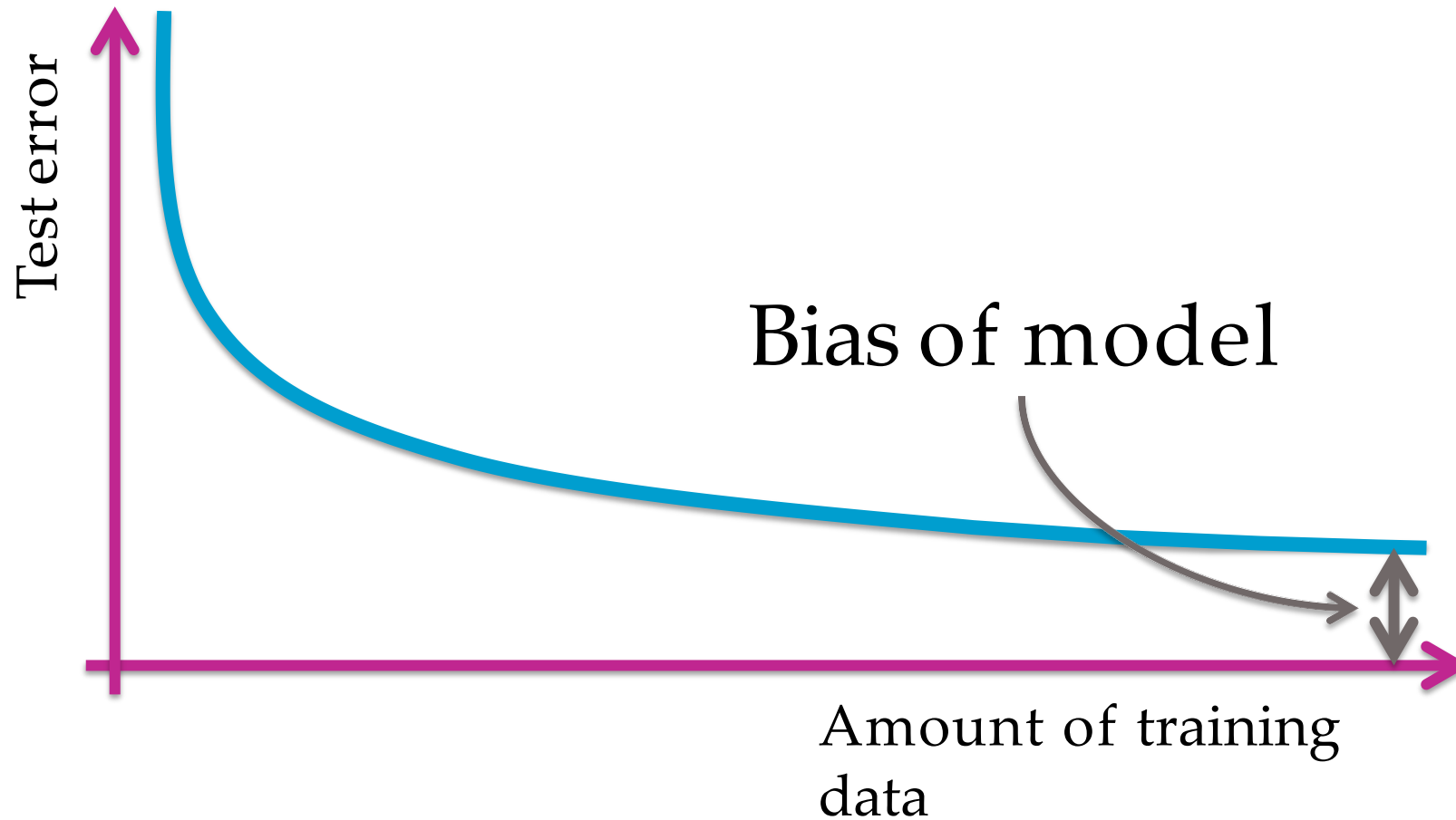
Learning curves



Learning curves

Is there a limit?

Yes, for most models...



Learning curves

More complex models tend to have less bias...

Sentiment classifier using single words can do OK, but...

Never classifies correctly: "The sushi was not good."

More complex model: consider pairs of words (bigrams)

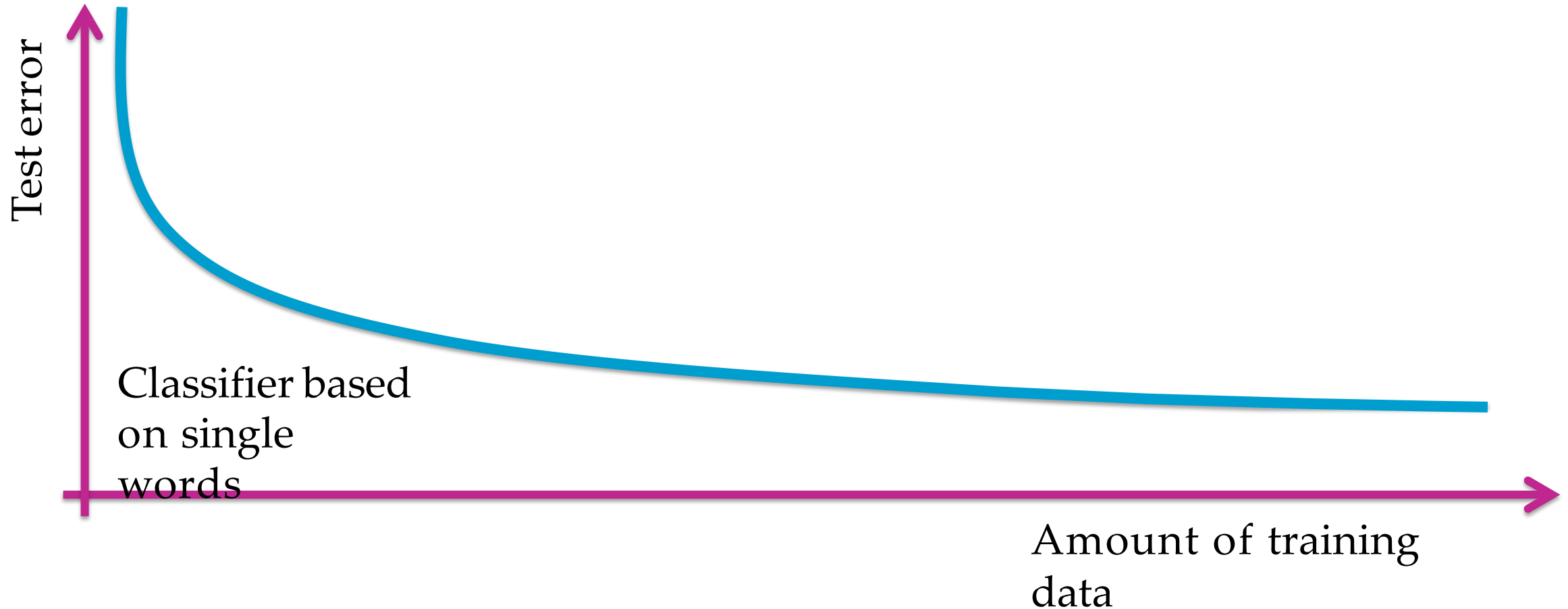
Word	Weight
good	+1.5
not good	-2.1

Less bias →

potentially more accurate, needs
more data to learn

Learning curves

Models with less bias tend to need more data to learn well, but do better with sufficient data



Class probabilities

How confident is your prediction?

- Thus far, we've outputted a prediction
- But, how sure are you about the prediction?



- *"The sushi & everything else
were awesome!"*



$$P(y=+ | x) = 0.99$$

Definite

- *"The sushi was good, the
service was OK."*



$$P(y=+ | x) = 0.55$$

Not sure

Many classifiers provide a confidence level:

$$P(y | x)$$

Output label



Input sentence



Extremely useful in practice

Summary of Classification

What you can do now...

- Identify a classification problem and some common applications
- Describe decision boundaries and linear classifiers
- Train a classifier
- Measure its error
 - Some rules of thumb for good accuracy
- Interpret the types of error associated with classification
- Describe the tradeoffs between model bias and data set size
- Use class probability to express degree of confidence in prediction

Python Implementation

Python Implementation

Count the words in sentences

```
from sklearn.feature_extraction.text import CountVectorizer
corpus = [
    'This is the first document.',
    'This document is the second document.',
    'And this is the third one.',
    'Is this the first document?',
]
vectorizer = CountVectorizer()
X = vectorizer.fit_transform(corpus)
vectorizer.get_feature_names_out()
>>>array(['and', 'document', 'first', 'is', 'one', 'second', 'the', 'third',
        'this'],...)
```

- Fit: Analyzes the text in the training data (X_train) to learn the vocabulary (i.e., all the unique words in the training set) and their corresponding feature indices.
- Transform: Converts the training text into a word count matrix based on the learned vocabulary.

```
print(X.toarray())
>>>[[0 1 1 1 0 0 1 0 1]
    [0 2 0 1 0 1 1 0 1]
    [1 0 0 1 1 0 1 1 1]
    [0 1 1 1 0 0 1 0 1]]
```

Think about it: How to train a model to predict sentiment?

Tutorial

